



Ariel Ramos Perez

786-803-7916 | arielrperez@gmail.com | [linkedin.com/in/ariel-ramos-perez](https://www.linkedin.com/in/ariel-ramos-perez) | github.com/rihhito | rihhito.github.io

EDUCATION

M.S. Computer Engineering, Robotics Concentration

Florida International University

Expected Graduation: Dec 2027

Miami, FL

B.S. Computer Science, Minor in Mathematical Sciences

Florida International University, Honors College

Jun 2022 – Dec 2025

Miami, FL

- GPA: 3.98 / 4.00

Relevant Coursework: Autonomous Systems & Controls, Robot Control, Advanced Design of Robots, Modeling & Simulation, Data Structures, Operating Systems, Neural Networks, Machine Learning, Artificial Intelligence, Random Signal Principles, Advanced Embedded Systems & IoT

EXPERIENCE

Software Engineering Intern

Savannah River National Laboratory

Jun 2026 – Aug 2026

Aiken, SC

- Trained a reinforcement learning policy in NVIDIA Isaac Sim/Isaac Lab that taught a UR-20 robotic arm to scrape dried radioactive waste from tank surfaces, delivering a working end-to-end prototype within the 10-week internship.
- Integrated NVIDIA Isaac ROS cuMotion and nvblox to reconstruct the workcell in 3D and generate collision-free motion plans, validating trajectories in simulation before hardware runs.
- Containerized the development environment with Docker on Linux and managed the codebase in Git, giving the team reproducible builds and simulation runs across shared GPU workstations.
- Integrated and calibrated ZED 3D and Intel RealSense D435 depth cameras, writing Python tooling to align point clouds and diagnose sensor dropouts during testing.

Student Researcher

Applied Research Center, Florida International University

Jul 2025 – Present

Miami, FL

- Develop ROS 2 nodes in C++ and Python for the autonomous navigation of a Clearpath Husky A300, using GPS as the primary odometry source.
- Fielded a Husky A200 carrying ground-penetrating radar and LiDAR to scan walls, floors, and ceilings, capturing subsurface data in areas unsafe for manual inspection.
- Built an interactive 3D visualization tool that renders raw GPR returns as navigable volumes, replacing manual inspection of 2D radargrams and cutting scan review time.

Course Assistant, Human-Computer Interaction

Florida International University

Jan 2025 – Apr 2025

Miami, FL

- Partnered with the instructor to design assignments and teaching materials, gave written feedback on student design and implementation work, and graded against a shared rubric to keep evaluation consistent.

PROJECTS

ArtLens: MET Insights | Swift, Python, MySQL, Streamlit, REST APIs

Aug 2024 – Dec 2024

- Built a web and iOS application delivering an interactive experience of the Metropolitan Museum of Art collection, contributing to both client-side and server-side code.
- Designed the MySQL schema and a Streamlit administrator dashboard for managing artwork records, user accounts, and recommendation data.
- Implemented the artwork recommendation feature in Swift/Xcode, connected the iOS client to the backend API, and coordinated the work over Git with branch-based code review.

TECHNICAL SKILLS

Languages: C++ (C++17/20), C, Python, Swift, SQL

Robotics & Simulation: ROS 2, Isaac Sim/Isaac Lab, Isaac ROS (cuMotion, nvblox), Unreal Engine, Unity

ML Frameworks: PyTorch, TensorFlow

Tools: Linux, Git, Docker, CMake

Spoken Languages: English, Spanish (fluent)